When Language Models Quote Uncertainty: Separating Reasoning Errors from Measurement Noise

Raghhav Malani
PES University, Bengaluru

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Abstract

A numerical answer with a stated uncertainty is useful only if the uncertainty is honest problem by problem. We study this with a generator of multi-step physics problems that knows each problem's true uncertainty $\sigma^\star$, sets measurement noise independently of chain depth, and builds counterfactual pairs that separate propagation from surface reading. Training Qwen2.5-Math-1.5B-Instruct against a proper scoring rule, we find three things. First, with sequence-level credit the stated σ collapses; giving its tokens their own advantage raises 1σ coverage from 10.8 % to 41.8 %, but σ still does not rank errors. Second, on the fine-tuned model and its terse format, the tested single-completion signals (stated σ, a hidden-state probe, self-verification) provide little within-problem discrimination between correct and incorrect answers; agreement among samples discriminates better, at 8.0 times the tokens. Third, trained on visible sensitivity derivations, the model states a σ that propagates measurement noise on fresh problems in a pre-registered test (slope +0.82, ideal one), yet its expected score is not shown to beat a root-sum-square heuristic: it closes 7 % of the gap to $\sigma^\star$, and the loss lies in derivatives of value-dependent relations. Learning how uncertainty propagates is not the same as forecasting it well. Results are for one small model on a synthetic benchmark; an exploratory replication on a second model family returned NO-GO.

1 Introduction

A language model that answers a numerical question can also state how uncertain the answer is, writing "$\mu \pm \sigma$" with a unit. Such a statement is useful only if it is honest problem by problem: wide when the answer is likely to be wrong or the inputs are noisy, narrow when neither holds. Pooled calibration does not establish this. A single scale factor applied after training can bring a model's intervals to nominal coverage on average while saying nothing about which problems are uncertain, so we treat such a factor as a baseline that every learned uncertainty must beat. We ask where a model's numeric uncertainty should come from, and whether training can make one stated σ track the true uncertainty of each problem.

Answering that needs problems whose correct uncertainty is known one by one, which real datasets do not provide. We build a procedural generator of multi-step physics problems (3 formula families, chains of up to 5 steps) in which each given can carry stated measurement noise of up to 5 %. The generator computes each problem's true predictive uncertainty $\sigma^\star$ by Monte Carlo through the chain, and it sets the noise independently of the chain's depth. A problem's error then has two separable sources: a mistake in the reasoning, which makes the answer wrong whatever the inputs, and noise in the givens, whose effect $\sigma^\star$ describes exactly. Counterfactual pairs of problems isolate the second. Within a pair, the displayed values and the multiset of noise levels are identical

and only the given carrying the high level changes, so a reader of surface features cannot move its σ, while an exact propagator moves it by the change in $\ln \sigma^\star$.

We train Qwen2.5-Math-1.5B-Instruct with supervised fine-tuning and reinforcement learning against a proper scoring rule, and report three findings, one per panel of Figure 1.

1. **Optimization (Section 6).** With sequence-level credit, a proper scoring rule did not stop the stated σ from collapsing towards zero. Giving the tokens of σ their own advantage raises 1σ coverage from 10.8 % to 41.8 %, but the wider σ still does not rank errors.
2. **Within-problem correctness signals (Section 7).** On the fine-tuned Qwen2.5-Math-1.5B-Instruct model and its terse response format, the tested single-completion uncertainty signals provide little within-problem discrimination between correct and incorrect answers; sample agreement provides stronger discrimination, at 8.0 times the tokens of one answer. This is specific to the evaluated checkpoint and elicitation conditions and does not establish that small language models generally lack accessible answer-correctness information.
3. **Propagation of measurement noise (Sections 8 and 9).** Trained on visible sensitivity derivations, the model states a σ that propagates measurement noise on fresh problems (a noise-allocation slope of +0.82 against an ideal of one), and through Jacobians outside the training range of trained chains (+0.99, although a propagator clamped to the training range already reaches +0.83). Yet its expected score is not shown to beat the root-sum-square heuristic: it closes 7 % of the gap between that heuristic and $\sigma^\star$. The loss sits in derivatives of value-dependent relations.

The central message is a tension: learning how uncertainty should propagate is not the same as forecasting it well. The confirmed propagation result and the unsupported score comparisons come from the same pre-registered run, and we report them with equal prominence.

Our contributions are narrow:

- a benchmark of physics problems with a known per-problem $\sigma^\star$, measurement noise set independently of depth, and counterfactual pair sets that separate propagation from surface reading (Section 3);
- an experimental separation of reasoning errors from measurement noise, with a within-problem test that holds difficulty fixed (Sections 7 and 8);
- the gap between propagation and forecast quality, located by a tail autopsy and by exact-component swaps (Section 8);
- a pre-registered protocol with a dated public decision log, in which amendments are labelled and negative results are reported with the same prominence as positive ones (Section 5).

Scoring a continuous answer with a CRPS reward is not new [20], and agreement among samples is a known confidence signal [27]; what is new here is a single stated σ evaluated against a known $\sigma^\star$. Every claim is scoped to 4-bit Qwen2.5-Math-1.5B-Instruct with LoRA, this synthetic benchmark and these training procedures; an exploratory one-seed replication on a second model family returned NO-GO (Section 9).

2 Related work

Calibration and verbalized confidence. Language models can estimate whether their own answers are correct, for example by stating the probability that a proposed answer is true [17]. For

models fine-tuned with human feedback, confidences stated in words are often better calibrated than the model's token probabilities [24], but stated confidences tend to be overconfident, and consistency among several sampled answers can help [28]. Language and vision-language models are mostly overconfident when they verbalize uncertainty, and vision-language models are poorly calibrated when they state a mean and standard deviation or an interval for a numerical answer [11]. Intervals that models construct around their own numerical estimates cover the truth far less often than their nominal level [6], including in quantitative forecasting [21]. These works evaluate pooled calibration against outcomes; because the right uncertainty of each question is unknown, they cannot test whether a stated interval tracks it. Our generator knows each problem's $\sigma^\star$, so we can ask that per problem, and our pair sets ask whether σ moves when only the source of noise moves.

Reinforcement learning with proper scoring rules. RLCR adds a Brier score on a stated confidence to a binary correctness reward, and proves that a bounded proper scoring rule in the reward yields answers that are both accurate and calibrated [3]. Rewarding Doubt trains a stated confidence on factual questions with the logarithmic scoring rule [1]. Both target the probability that an answer is correct, not a distribution over a continuous quantity. Confidence-aware RL is limited by the model's confidence prior before RL, which can concentrate on a few high values and persist through training [26]; under sequence-level credit we observe a related failure, a stated σ collapsing towards zero (Section 6). For LLM regression, DAR scores the empirical distribution of several sampled predictions and credits each prediction by its leave-one-out contribution [20]. A CRPS-type reward is therefore not our contribution: in DAR the uncertainty is the spread of rollouts, whereas we train a single stated σ and score it against a known $\sigma^\star$.

Sampling, self-verification and executable computation. Self-consistency samples several reasoning paths and takes the most consistent answer [27]; we use the agreement among samples as a correctness signal and report its cost beside its discrimination. Without external feedback, models struggle to correct their own reasoning, and their performance can even degrade when they try [15]; our self-verification probe is consistent with this, since the model accepts almost all of its own answers. Program-aided reasoning delegates the computation to an interpreter [2, 7]. Our MC-B applies that idea to uncertainty: the model writes the problem's program and the program is run on perturbed inputs. These works limit MC-B's novelty, and MC-B failed (Section 8).

Uncertainty propagation and proper scoring. Our $\sigma^\star$ follows the metrological practice of propagating distributions through a measurement model by Monte Carlo [16], and the root-sum-square heuristic is the first-order rule for a product of first powers. Our reward and evaluation score, the CRPS in log space, is strictly proper for log-normal beliefs in the sense of Gneiting and Raftery [8], so an honest σ is optimal in expectation; whether optimization finds it is the question of Section 6.

3 Setup

3.1 A generator of physics problems with known uncertainty

The benchmark is a procedural generator of multi-step physics word problems (generator version 2.0.0). It has 3 formula families (kinematics, dynamics, energy), each one physical scenario with named, dimensioned quantities and a set of relations written in SymPy [19]: 12, 15 and 10 relations respectively. A problem is a linear chain through its family's formula graph. The generator samples

a consistent world (basis quantities log-uniform within physical ranges, derived quantities checked against plausibility bounds), picks a chain, shows the chain's inputs (the *givens*) and asks for its output (the *target*). The depth of a chain, from one step up to 5, is its difficulty dial. Depth is honest: the target's shortest derivation from the givens, over all relations of the family, has exactly the chain's length, and no given can be derived from the others. Depth 5 is never used for training. It tests extrapolation in chain length within the same families, not transfer to new physical laws.

Each problem's text is randomized: phrasing templates, display units with SI prefixes and non-SI units, and 0–2 distractor quantities that the answer does not depend on. Chains whose output amplifies relative errors in a given by more than a factor of 5 (the largest $|\partial \ln(\text{output})/\partial \ln(\text{given})|$) are rejected. The ground truth is computed in SI units with dimension checks at every step, then converted to the requested unit. Every problem is reproducible from its generator version and seed.

3.2 Measurement noise and the true uncertainty $\sigma^\star$

In the measurement-noise mode, each noisy given is shown as a measurement $x_{\text{meas}} = x_{\text{true}}(1+e)$, $e \sim \mathcal{N}(0, s^2)$, with 5 significant figures, so that rounding is negligible against the noise, and a stated relative uncertainty "$\pm\, s$"; a value shown without one is exact. The truth is the noise-free chain evaluated on the unrounded true values. A problem without noise shows 3 significant figures, and its truth is computed from the displayed values. The pair sets of Section 3.5 use the levels 1 %, 2 %, 5 %, and the training data of Section 4.4 draws a level for each given independently from 0 %, 1 %, 2 %, 5 %, where a level of zero means an exact given.

Because the generator knows the measurement model, it knows each problem's correct uncertainty. Given the displayed values, the clean value of each noisy given is drawn as $x = x_{\text{meas}}/(1+e')$, $e' \sim \mathcal{N}(0, s^2)$, with exact givens and constants held fixed; 4,000 such draws are pushed through the chain, as in Monte Carlo propagation of distributions [16]. The target $\sigma^\star$ is the scale of the Gaussian in log space, centred at $\ln f(x_{\text{meas}})$, that minimizes the expected CRPS against these draws:

$$\sigma^\star = \arg\min_{\sigma>0} \; \mathbb{E}_x\Big[\mathrm{CRPS}\Big(\mathcal{N}(\ln f(x_{\text{meas}}), \sigma^2),\, \ln f(x)\Big)\Big], \tag{1}$$

where f is the chain and $f(x_{\text{meas}})$ is what a correct solver computes from the displayed values. The standard deviation of the log draws and the first-order (delta-method) value are stored as well.

Before any model was evaluated on noisy problems, the generator had to pass a self-consistency gate: intervals $\ln f(x_{\text{meas}}) \pm k\sigma^\star$ must cover the clean truth at their nominal rates. On the final run, with 10,005 noisy problems in 15 depth-by-level cells of 667 each, pooled coverage was 68.32 % at $k = 1$ (nominal 68.27 %, Kupiec [18] $p = 0.92$) and 95.71 % at $k = 2$ (nominal 95.45 %, $p = 0.20$). No cell was rejected after Holm correction, and the probability integral transform of the truth was uniform (Kolmogorov–Smirnov $p = 0.98$). The 500 noise-free control problems were exact. The gate failed twice before this run. The first run failed on one cell at the highest noise level, after which the noise levels were capped; that cap was chosen after seeing the result. The second failed only on a rule that, as written, tolerated no cell outside its interval; it was replaced by a miss-count test before the final run. Both changes are listed in Table 2.

3.3 Output format and verifier

A completion may reason freely, then reports the requested intermediate values in a `<steps>` block and the answer as `<answer>`$\mu \pm \sigma$ *unit*`</answer>`. A completion that does not parse, or whose unit has the wrong dimension, scores the rule's floor (-5); an answer in another unit of the right dimension is converted.

The scoring rule, $\mathrm{crps}_{\log}$, scores the log of the truth y against $\mathcal{N}(\ln\mu, (\sigma/\mu)^2)$:

$$\mathrm{crps}_{\log}(\mu, \sigma; y) = -\,s\left[z\,(2\Phi(z)-1) + 2\varphi(z) - \tfrac{1}{\sqrt{\pi}}\right], \qquad z = \frac{\ln y - \ln\mu}{s}, \quad s = \max(\sigma/\mu,\ \epsilon)\,, \quad (2)$$

the Gaussian CRPS in closed form [9], clipped below at the floor. It is strictly proper for log-normal beliefs [8] and scale-free. The relative CRPS of the original specification divides by the truth, which reweights outcomes and makes it only approximately proper; it remains an ablation option. The floor ϵ is 1×10^{-6} (verifier 1.1.0): an earlier, larger floor made the reward flat in σ exactly where an overconfident policy needed a gradient. During reinforcement learning the reward adds 0.2 times the share of requested intermediates within 2 % of their values; this weight is checked to be too small to ever outweigh the final score. An answer is *accurate* if it is within 1 % of the truth.

On noisy problems, an answer is *computation-correct* if $|\ln\mu - \ln f(x_{\mathrm{meas}})| \le 1 \times 10^{-3}$, that is, if it is what a correct solver computes from the displayed values. Comparisons of a stated σ with $\sigma^\star$ that require a correct answer use this subset.

3.4 The oracle-centred expected score

A single realized truth says little about whether a stated σ is right: a σ that is 2 times too small scores better than $\sigma^\star$ against the realized truth on 44 % of problems. Because the generator knows the predictive distribution, we also score a stated relative σ by its *expected* $\mathrm{crps}_{\log}$ over 4,000 fresh predictive draws, centred at $f(x_{\mathrm{meas}})$. Per-problem differences between sources then have 5–23 times less variance than the realized ones. Expected-score differences are reported with their intervals and, where useful, as a share of the gap between the noise heuristic RSS (Section 4.5) and $\sigma^\star$, which is 0.00296 on the frozen confirmatory set.

3.5 Counterfactual pair sets

A model can state a plausible σ without propagating anything, by learning which templates, depths and stated percentages go with which uncertainty. To separate propagation from such a lookup, we evaluate on matched pairs of problems that differ in exactly one respect (Table 1). The statistic is the within-pair slope: the ordinary least-squares slope, with an intercept, of $\Delta\ln\sigma_{\mathrm{stated}}$ on $\Delta\ln\sigma^\star$ across pairs, where Δ is the difference between a pair's two members. An exact propagator has slope one; a reader of features that do not differ within a pair has slope zero.

A6 (value changes). One chain template, two worlds: of 24 candidate worlds, the two with the lowest and highest delta-method $\sigma^\star$, kept if their ratio is at least 2.0. Template, phrasing, noise level and the standardized measurement errors are shared. Power-law chains cannot form A6 pairs, since their relative $\sigma^\star$ does not depend on the values.

A6b (invariance). Power-law templates only, two worlds whose targets differ by a factor of at least 2.0, with relative $\sigma^\star$ checked equal by Monte Carlo (to within 1×10^{-3} in log). Here the slope is taken on $\Delta\ln y$ and must be zero: a model whose relative σ grows with the size of the answer fails it.

A6c (noise allocation). One chain, one world and identical displayed values. The two members differ only in which given carries the higher of two noise levels: in one member it sits where it moves $\sigma^\star$ least, in the other where it moves it most (ratio at least 1.5). Both members show the same multiset of percentages, so no reader of the problem's surface can tell them apart.

Each development set has 15 depth-by-level cells of 34 pairs. The frozen confirmatory set (Section 5.2) repeats the three constructions with fresh seeds, 1,020 problems each.

Table 1: The counterfactual pair sets and what each isolates.

set	changes within a pair	held fixed	regressor	ideal	problems
A6	the world (values); same template	template, phrasing, level, z	$\Delta \ln \sigma^\star$	1	1,020 / 1,020
A6b	the target's magnitude ($\geq \times 2$); power-law template	relative $\sigma^\star$ (MC-checked)	$\Delta \ln y$	0	1,020 / 1,020
A6c	which given carries the high level	displayed values, level multiset	$\Delta \ln \sigma^\star$	1	1,020 / 1,020
J sensitivity	as A6c, on a Jacobian outside the training range	as A6c	$\Delta \ln \sigma^\star$	1	816
J invariance	a dimensional rescaling (target $\times 2$–8)	every log-sensitivity, so relative $\sigma^\star$	$\Delta \ln y$	0	816

Ideal: the within-pair OLS slope of $\Delta \ln$(stated σ) on the regressor for an exact propagator; a reader of surface features that are identical within a pair scores 0 on A6c and J. Problems: development / frozen confirmatory set (A6, A6b, A6c; D113), or the frozen J set (D147). Sources: the sets' manifests.

Jacobians outside the training range (the J set). A model trained on a chain might have learned that chain's typical sensitivities rather than computing them. The J set therefore uses only chains that occur in the training data, and moves them to Jacobians they never had there. For each of the 1,149 chains in the 3,000 training problems we record the range of every given's log-sensitivity $\partial \ln f / \partial \ln x$ at the displayed values. A J world has at least one given whose sensitivity lies outside that range by at least 0.1, while every displayed and computed value stays inside the range that quantity had in training; the first such world of up to 96 draws is used. *J sensitivity pairs* apply the A6c construction to J worlds. *J invariance pairs* pair a J world with a copy rescaled along a dimensional direction that leaves every physical constant unchanged, which keeps every log-sensitivity, and so the relative $\sigma^\star$, identical while the target changes by a factor of 2 to 8; A6b cannot be used here because its power-law Jacobians are constant. The J set has 816 sensitivity and 816 invariance problems at the trained depths (up to 4). Across the sensitivity pairs, the most out-of-range sensitivity lies a median of 0.38 outside its chain's range, and the median chain was seen 3 times in training.

4 Methods

4.1 Model and training infrastructure

All models are Qwen2.5-Math-1.5B-Instruct [29], loaded with 4-bit NF4 quantization [5] from a pinned snapshot and trained with LoRA adapters [14] of rank 16 ($\alpha = 16$) on all 7 attention and MLP projections, using Unsloth [12] and TRL [25]. Adapters are evaluated as trained: unmerged, on the same quantized base. An early evaluation path merged reinforcement-learning adapters into a bf16 copy of the model, which erased most of the update; those results are superseded and kept only as labelled records. A cross-model pilot repeats the D-pos-v2 and B-v2 recipes of Section 4.4 on Llama-3.2-3B-Instruct [10].

4.2 Supervised warm-up

Reinforcement learning starts from a supervised warm-up on 3,000 generator solutions written in the output format: the needed SI values, one line per chain step with the substituted numbers, the `<steps>` block and the answer. The σ in these targets is a fixed 2 % of the value, and its tokens are masked from the loss, so the warm-up teaches the format and the computation but never a particular uncertainty. It runs for 30 minutes at learning rate 1×10^{-4}.

4.3 Reinforcement learning with a proper scoring rule

We train with GRPO [23]: for each of 4 problems per step the policy samples 8 completions at temperature 1.0, each completion is scored by the verifier (Section 3.3), and the reward is normalized within its group. There is no KL penalty ($\beta = 0$); the learning rate is 5×10^{-6} and completions are capped at 512 tokens. Problems are drawn by a curriculum over (family, depth) cells that tracks each cell's pass rate p with an exponential moving average (rate 0.1) and samples cells in proportion to $p(1-p) + 0.02$, the weight that favours groups with reward variance. Runs are compared at a matched step (287).

Split credit. With sequence-level credit every token of a completion receives the same advantage, so the tokens of σ are rewarded for the answer's correctness rather than for their own calibration. Split credit decomposes the verifier's reward r into two parts that sum to it,

$$r_{\text{answer}} = \text{crps}_{\log}(\mu, \sigma_{\text{ref}}) + w\, r_{\text{steps}}, \qquad r_\sigma = \text{crps}_{\log}(\mu, \sigma) - \text{crps}_{\log}(\mu, \sigma_{\text{ref}}), \tag{3}$$

with a fixed reference σ_{ref} of 1 % of μ and the step weight w of Section 3.3. Each part is normalized within its group. The tokens of the σ number, located by parsing, receive the advantage of r_σ; every other token receives that of r_{answer}. A failed completion keeps its whole reward in r_{answer}. The verifier, the logged reward and every hyperparameter are those of the uniform-credit run; split credit and uniform credit are each trained with 3 seeds.

σ-only reinforcement learning (E1). To isolate credit assignment completely, E1 holds the answer fixed. Each training prefix is a warm-up model's own completion truncated right after the "$\pm$" of its answer, and the policy samples only the remaining σ suffix (at most 24 tokens), so that within a group only σ varies. E1 trains for 300 steps at learning rate 5×10^{-6}; the E1d pilot repeats it at 5×10^{-5}, and is treated as exploratory.

4.4 Explicit propagation: D-pos-v2 and its implicit control B-v2

Supervising $\sigma^\star$ directly, rewarding σ alone, and training an explicit propagation line on data without value-dependent derivatives all failed to make the stated σ propagate noise. The data of this section contain the behaviour. They reuse 3,000 training problems with mixed noise: each given's level is drawn independently from 0 %, 1 %, 2 %, 5 %, and 2,067 problems mix two or more noisy levels. Both arms see the same problems.

D-pos-v2 (explicit). After the solution, the target completion writes one line per noisy given x_i: the given's total log-sensitivity $S_i = \partial \ln f / \partial \ln x_i$ as a symbolic expression derived by the chain rule over the steps, the same expression with the solution's values substituted, and its value to 3

significant figures. A power-law term is stated as its exponent. A propagation line then combines the terms with the stated levels s_i,

$$\sigma_{\text{rel}} = \sqrt{\textstyle\sum_i (S_i\, s_i)^2} = X, \tag{4}$$

and the answer states $\sigma = X\,|\mu|$. Every token is trained, including the sensitivities, the line and σ. We read three values of σ from each D-pos-v2 completion: (a) the answer's relative σ; (b) the line's stated X; (c) the exact value of Equation (4) computed from the line's own written terms. Comparing them locates a failure in the propagation itself, the arithmetic of the line, or the copy into the answer.

B-v2 (implicit). The same problems and solutions with no sensitivity lines and no propagation line; $\sigma^\star$ itself is written after "±" and trained. B-v2 can learn propagation only implicitly.

Training and evaluation. Each arm trains a fresh adapter on the pinned base for 2 epochs (750 steps) at learning rate 1×10^{-4} and effective batch 8, with 3 seeds per arm. The seed changes the adapter's initialization and the data order, not the data. Evaluation is greedy, with up to 1,280 new tokens, enough for the longest derivations.

4.5 Reference uncertainties

Every stated σ is compared with three references that need no model. $\sigma^\star$ is the generator's own value (Equation (1)). RSS is the root sum of squares of the stated levels of the givens, $\sqrt{\sum_i s_i^2}$: correct for a product of first powers, and blind to the values. X1 is a shallow text model of $\sigma^\star$: gradient-boosted trees on the problem text's tokens, the stated levels and the displayed magnitudes, fitted on generator problems disjoint from every evaluation set. It is refit exactly and used only if it reproduces its published results. On the J set we add a fourth, *clamped* reference: the first-order σ with every sensitivity clipped into its chain's training range, which is what a model that propagates exactly but only within its training range would state.

4.6 Executing instead of predicting: MC-A and MC-B

MC-A estimates σ by re-asking the model. For each problem it builds 8 perturbed copies in which every noisy given is redrawn from the generator's own predictive distribution, $x = x_{\text{meas}}/(1 + sz)$ with $z \sim \mathcal{N}(0, 1)$, and shown in its displayed unit; B-v2 answers each copy. Its σ is the spread of $\ln \mu$ over the copies: the standard deviation (raw), or 1.4826 times the median absolute deviation over the copies whose solution steps match the unperturbed one (robust). The same perturbations pushed through the exact chain give an oracle that targets $\sigma^\star$.

MC-B asks the model for the computation instead of the answer, in the spirit of program-aided reasoning [2, 7]. The base model, without an adapter, writes a Python function `solve` whose parameters are the givens' own symbols; 2 worked examples from a disjoint seed range at depth 2 show the format, and generation is greedy with up to 640 new tokens. A sandbox checks the program's syntax tree (only the `math` module, no dynamic code), calls it on the displayed values and on MC-A's perturbations, and takes the spread of the logarithms of its outputs as σ. A failing program is classified as an interface failure (no program, wrong signature, rejected), a runtime failure, or a semantic failure (it runs but computes the wrong chain).

5 Evaluation protocol

5.1 Pre-registration and the decision record

The project's hypotheses were pre-registered in a file written before any result and never edited afterwards. Every experiment after that was specified in a dated entry of a public decision log, pushed to the repository before its data existed; the commit is the timestamp, and the few amendments made after a result are labelled (Section 5.4). The log has 153 entries. Each pre-registration states the hypothesis, the estimator, the decision rule and what would count as failure, and labels the analysis confirmatory or exploratory. Decisions taken during unattended runs were logged in the same way, with the alternatives considered. Results are recorded as numbered findings (39 so far), each citing its entry, and a negative result is reported with the same prominence as a positive one. Pre-registered hypotheses that were not run are declared as such rather than dropped. A chronological timeline of every pre-registration and amendment, with its status, accompanies the paper.

5.2 Frozen evaluation sets

The development pair sets of Section 3.5 were used to build and debug every analysis. Confirmatory claims rest on frozen sets built with fresh seeds after their analysis was specified: the R4-5 set (A6, A6b and A6c, 1,020 problems each) and the J set. Each frozen file has a SHA-256 manifest; the evaluation loader refuses a frozen file unless the run sets an explicit confirmatory flag, and regenerates every problem from its seed and compares it before any generation. No model saw a frozen set before the confirmatory run that its entry registered. While those runs were in progress, interim reports gave only operational counts (completions, truncation, parse failures, memory backoffs), and each confirmatory analysis ran once, after all of its evaluations existed.

Every completion is logged with its problem, verifier breakdown and provenance, and replaying a log regenerates each problem from its seed and re-scores the completion exactly. Every adapter carries a fingerprint recorded at the end of training, and an evaluation reloads it and checks the fingerprint before generating. Long evaluations write each finished batch to disk under a key of the model, adapter, prompts and generation settings, so an interrupted greedy evaluation resumes without changing any completion.

5.3 Statistics

Pair slopes. For each pair set the statistic is the within-pair OLS slope with an intercept (Section 3.5): of $\Delta \ln \sigma_{\text{stated}}$ on $\Delta \ln \sigma^{\star}$ for A6, A6c and the J sensitivity pairs, and on $\Delta \ln y$ for A6b and the J invariance pairs. A stated σ of zero enters at a relative floor of 1×10^{-6}. For an answer's σ, the noise-allocation slopes use pairs whose two members are both answered, and the invariance slopes use pairs whose two members are both computation-correct, so that a wrong answer cannot pass as invariance; values read from a propagation line use every pair.

Uncertainty over pairs and seeds. Confidence intervals come from a bootstrap over pairs with 10,000 resamples. With several seeds, each resample draws the pairs once and recomputes every seed's statistic on them; the estimate is the mean of the per-seed statistics [4]. One-sided p-values are the share of resamples at or below zero. Differences in expected score (Section 3.4) are averaged over seeds per problem and bootstrapped with pairs as clusters. Within-problem comparisons of correctness signals are clustered over problems. Neither pairs nor seeds that share problems are ever treated as independent.

The propagation criterion. A σ source propagates if three conditions hold: the noise-allocation slope is positive (C1: the lower end of its 90 % interval is above zero); it is at least 0.5, a practical threshold (C2); and the invariance slope is equivalent to zero, with its 90 % interval inside ± 0.3, a two one-sided tests criterion [22] (C3). The primary hypotheses of the confirmatory runs (P on the R4-5 set, P_J on the J set) apply this criterion to D-pos-v2's answer σ, with C1 replaced by a one-sided test that is Holm-adjusted within its family.

Multiplicity. Each confirmatory run fixes its family of tests in advance (K1–K8 on the R4-5 set, J1–J6 on the J set), and p-values are Holm-adjusted [13] within it at level 0.05. Subgroup estimates (by depth, chain type or distance from the training range) are reported with intervals but are outside the family, and no confirmatory claim rests on them. A score difference that is not significant is reported as not rejected, never as a tie.

5.4 Deviations

Table 2 lists every amendment to a pre-registered plan and every deviation that touched a registered analysis, with when it was made relative to the data it affects. Most were made before any affected result existed. Those made after a result are labelled as such wherever the result is reported.

Table 2: Amendments and deviations from the pre-registered plans.

entry	date	what	timing
D47	10-14 10:41	P's pre-registered sanity check failed; exploratory baselines added afterwards	amendment, **after** P's result
D49	10-14 11:54	Two post-hoc calibration baselines; RL calibration judged against the text-feature one	amendment, before RL calibration results
D50	10-14 11:59	Review round 1, A1–A8: two estimands; the σ^* definition; the generator self-consistency gate (A3); statistics; nested A5 baselines and family N …	amendment, before any result it affects
D54	10-14 12:43	Noise capped at 5%; 10% becomes a heavy-tail regime	amendment, **after** the gate failed (#9)
D57	10-14 12:56	A3's cell-share rule replaced by a miss-count test; the next rerun is FINAL	amendment, **after** #11, before the final rerun
D61	10-16 14:34	A6 slope labels ("partial / full propagation")	amendment, before any A6 model result
D62	10-16 19:12	Split-credit seeds 1 and 2; the seed-mean S1–S3 test; greedy vs sampled σ	amendment to family S, before seed-1/2 data
D64	10-17 21:57	The label "anti-propagation" for a CI below 0	amendment, before any A6 model result
D67	10-18 00:06	The bf16 merge was a bug: corrected (unmerged, 4-bit) evals become PRIMARY for families S, N, E; merged results kept, labelled superseded	amendment, before any corrected result
D69	10-18 09:48	Post-hoc baselines re-fitted on 4-bit SFT calibration completions	amendment, before any such completion
D71	10-18 09:50	Review round 2, R2-1–R2-7: eval hygiene (reload fingerprint), SFT revalidation, seed-level statistics (R2-4), uniform-credit seeds (R2-6), …	amendment, before any result it affects
D77	10-19 10:31	R2-1: below the log-prob floor, the σ-token criterion decides; the check is enforced	amendment (future fingerprints only)
D91	10-20 23:22	X4: a direct mode, a perturbed control, the cluster bootstrap, token cost	amendment, before any X4 data
D92	10-20 23:38	X2 robustness checks, defined before computing	analysis plan, **after** X2's result

entry	date	what	timing
D93	10-20 23:45	Review round 3, R3-1–R3-10: X4 additions; X2 robustness; optimization pilots; Phase 12 scope = aleatoric propagation (R3-4); A6c (R3-5); …	amendment, before any result it affects
D112	10-25 17:13	Review round 4, R4-1–R4-9: D-pos-v2 vs B-v2 on mixed-level data, MC-A, the frozen set, framing (R4-8)	amendment, before any result it affects
D114	10-25 17:31	Two deviations found while building: eval budget 1,280 tokens; 6 rows dropped	deviation, before any R4 output
D116	10-25 19:07	D-pos-v2's eval OOM'd in its backoff; fixed and rerun at batch 16	deviation, before any completion was scored
D122	10-26 07:52	MC-B: two deviations found while building	deviation, before any output
D125	10-26 08:34	D118 §6 amended: the primary P is D113's full criterion (K1 ∧ C2 ∧ C3)	amendment, before any confirmatory result
D131	10-28 02:52	D-pos-v2 seed 1's eval hung after a GPU driver reset; rerun on identical code	deviation (operational), before the analysis
D133	10-28 10:04	Review round 5, R5-1–R5-10: framing; tail autopsy (R5-2); hybrid oracle (R5-3); RSS fallback (R5-4); cross-model replication (R5-5); cuts (R5-7); …	amendment, **after** #32 / #33, before anything it asks for
D135	10-28 10:24	R5-4's RSS-fallback bounds, σ/RSS ∈ [0.333, 5], chosen on the development sets and frozen before one application to the confirmatory logs	analysis plan; grid designed **after** seeing #35
D138	10-28 18:46	R2-1's fingerprint for the Llama base records at sampling batch 4	deviation (implementation), before any pilot output
D139	10-29 00:13	D137's go rule runs as soon as the D-pos-v2 eval exists (same code, same logs); SFT seed 1 only on GO	amendment (timing), before the go rule runs

Every amendment and deviation in the decision timeline (dates IST, 2026), with when it was made relative to the data it affects. Source: docs/decision_timeline.md.

6 Optimization: σ rides on the answer's advantage

The failure. This section and the next use the frozen noise-free evaluation set of 400 problems, with every adapter evaluated as trained (Section 4). With sequence-level credit, the tokens of σ share the answer's advantage, and the proper scoring rule did not keep the stated σ honest. Trained beyond the matched step to step 759, the uniform-credit policy states $\sigma = 0$ on 97.5 % of the evaluation answers, and its 1σ coverage falls to 2.8 %. The score does not favour this, since it is proper; the collapse is a property of the optimization. At the matched step (287) it is partial: across the three uniform-credit seeds, $\sigma = 0$ on 17.1 %–23.1 % of answers and the median relative σ is 0.041 %–0.057 %. The warm-up model's own σ is as narrow: a single scale factor fitted on the calibration split multiplies it by 22.1 to maximize the score, and would need 306 to reach nominal 1σ coverage there.

Split credit fixes coverage. Split credit gives the tokens of σ an advantage of their own (Section 4.3). With 3 seeds per arm, it raises mean 1σ coverage from 10.8 % under uniform credit to 41.8 % (Figure 1A), a difference of +31.0 percentage points [95 % CI +26.8, +35.2]; the warm-up model alone covers 12.0 %. Every split seed beats every uniform seed (9 of 9 cross-seed pairs; 41.2 %–42.5 % against 10.5 %–11.5 %). The collapse disappears: split credit states $\sigma = 0$ on 0.0 %–0.3 % of answers, with a median relative σ of 1.59 %–1.64 %. Accuracy holds (37.7 % against 37.2 %), so the pre-registered tests S1a, S1b and S3 are supported. Coverage nonetheless stays far below the nominal 68.27 %.

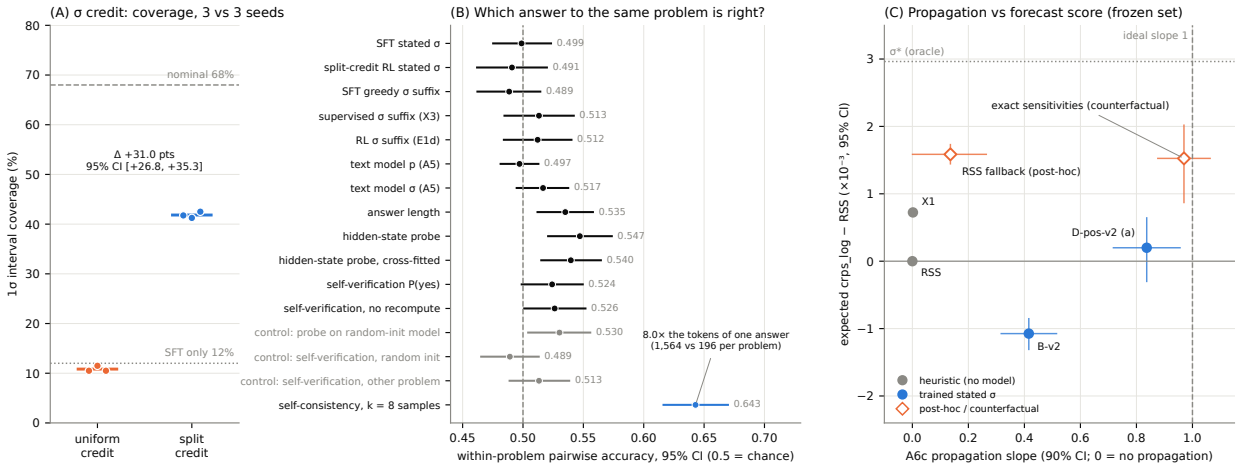


Figure 1: The three findings. (A) Uniform versus split credit: 1σ coverage per seed. (B) Within-problem pairwise accuracy of every correctness readout measured, against self-consistency. (C) On the frozen set, the noise-allocation slope against the expected-score difference from RSS. Every point is read from a committed result file; the sidecar `headline_data.json` traces each one.

It does not fix ranking. The wider σ says almost nothing about which answers are wrong. Its Spearman correlation with the error is 0.061 against uniform credit's 0.012, a difference of $+0.048$ [95 % CI -0.085, $+0.178$] with one-sided $p = 0.24$, so S2 is not supported. Every split seed's correlation (0.03–0.08) exceeds every uniform seed's (9 of 9 cross-seed pairs), but the values are tiny, and three seeds per arm cannot make this a seed-level significance claim. We judge RL calibration against text baselines (A5) that read the problem, the answer and its reasoning and predict the error of the same answers. Their σ correlates with the error at 0.64–0.68 per seed, and split credit's σ loses 9 of 9 pre-registered comparisons with it (family N: coverage error, correlation and score, per seed).

Learning σ with the answer fixed (exploratory). E1 takes credit assignment out of the question: the answer is fixed and only the σ suffix is sampled (Section 4.3). At learning rate 5×10^{-6}, E1 widens σ (1σ coverage 48.5 %), but its correlation with the error stays at 0.01; scaling the advantages by the batch instead of the group standard deviation (E1c) leaves it at 0.002. At 5×10^{-5} (E1d) the correlation rises to 0.586, $+0.573$ [95 % CI $+0.465$, $+0.680$] above E1's, the level that a supervised student of the A5 text baseline reaches on the same base and adapter (X3, 0.575). E1d's σ stays narrow (median 0.95 %, 1σ coverage 29.8 %), and its mean $\mathrm{crps}_{\log}$, though above that of the warm-up model's σ ($+0.010$ [95 % CI $+0.008$, $+0.012$]), stays below the A5 baseline's (-0.023 [95 % CI -0.032, -0.016]). We treat this as exploratory and scoped to the frozen-answer setting: it is a single-seed pilot on noise-free problems, the suffix is at most 24 tokens, and the learning rate is untested on whole completions, where the answer moves too. Its correlation also ranks problems, not answers: within a problem, E1d's σ is at chance (Section 7).

7 Within-problem correctness signals on the fine-tuned model

On the fine-tuned Qwen2.5-Math-1.5B-Instruct model and its terse response format, the tested single-completion uncertainty signals provide little within-problem discrimination between correct and incorrect answers; sample agreement provides stronger discrimination. This is specific to the

evaluated checkpoint and elicitation conditions and does not establish that small language models generally lack accessible answer-correctness information.

Holding difficulty fixed. A correlation between σ and the error across problems can come from difficulty alone: a readout that knows which problems are hard ranks them without knowing whether a given answer is right. To hold difficulty fixed, we sample $k = 8$ answers from the warm-up model for each of 1,200 problems and keep the 516 problems that have both right and wrong answers, 5,390 right–wrong pairs in all. A readout's pairwise accuracy on a problem is the share of its pairs in which the right answer gets more confidence (ties count half); we average it over problems and bootstrap 95 % intervals over problems. Chance is one half.

Single completions. Every readout of a single completion's own uncertainty lies in the range 0.491–0.524 (Figure 1B). The warm-up model's stated σ scores 0.499 [0.474, 0.524] and split credit's 0.491 [0.461, 0.521]; the A5 text baseline's probability of correctness 0.497 [0.481, 0.514] and its σ 0.517 [0.494, 0.538]; E1d's RL σ suffix 0.512 [0.483, 0.541] and X3's supervised suffix 0.513 [0.484, 0.543]. E1d and X3 agree to the third decimal: RL with clean credit and supervised distillation of the text baseline land in the same place. Answer length alone reaches 0.535 [0.511, 0.559]. A linear probe on the hidden state at the "±" token reaches 0.547 [0.520, 0.574], but the same probe on a randomly initialized copy of the architecture, reading the same tokens, reaches 0.530 [0.503, 0.557]. The difference, +0.017 [95 % CI −0.015, +0.050], does not exclude zero, so the probe does not show that the model represents its own error; nor is it distinguishable from length (−0.012 [95 % CI −0.045, +0.021]).

Self-verification. Asked to check its own answer, the model writes a verdict on 98 % of answers, and the verdict is "yes" on 99.9 % of the correct answers and 99.8 % of the incorrect ones. Its renormalized probability of "yes" is slightly higher on the incorrect answers (0.820 against 0.804). Within problems it reaches 0.524 [0.498, 0.550], which is not Holm-supported (one-sided $p = 0.033$ against a threshold of 0.0125). The check is short (a median of 13 generated tokens) and behaves like a test of whether the final line matches the steps: multiplying the final value by 1.25 lowers $P(\text{yes})$ on right and wrong answers alike, and the ranking does not move (0.524). Across problems it is anti-informative, with an AUROC of 0.469 [0.442, 0.496] for correctness.

Sample agreement, at a cost. Scoring each answer by the share of the problem's other samples that agree with it within 1 % gives the one strong within-problem signal: 0.643 [0.616, 0.671], +0.096 [95 % CI +0.058, +0.133] above the probe; E1d's difference from it is −0.131 [95 % CI −0.170, −0.091]. It needs the other samples. At 1,564 generated tokens per problem against 196 for one answer, it costs 8.0 times the tokens (6.07 against 0.76 GPU-seconds per problem; Table 3), and the cost grows linearly with k. The same trade-off appears on A6c for MC-A's spread over re-asked, perturbed copies (Section 8): at $k = 8$ its AUROC for a wrong answer is 0.745, against 0.666 for the best signal that needs no extra generation, at 1,984 extra tokens per problem (Table 4).

Across problems is a different question. Across all the sampled answers, E1d's and X3's σ correlate with the error at 0.350 and 0.346 (on the evaluation set, 0.586 and 0.575). Both rank which problems are hard; neither ranks the answers to one problem.

Table 3: Correctness signals against their inference cost.

readout	pairwise accuracy [95% CI]	tokens / problem
self-consistency (agreement of the other samples)	0.643 [0.616, 0.671]	1,564
hidden-state probe	0.547 [0.520, 0.574]	196
the same probe, random-init model	0.530 [0.503, 0.557]	196
completion length	0.535 [0.511, 0.559]	196
self-verification P(yes) (X4)	0.524 [0.498, 0.550]	196
stated σ (SFT)	0.499 [0.474, 0.524]	196

Within-problem pairwise accuracy (a right answer ranked above a wrong one to the same problem; 516 problems, 95% CIs clustered over problems) against generated tokens per problem (k = 8 samples for self-consistency). Source: results/r5_diag/cost.json (FINDINGS #22, #37).

Table 4: MC-A's spread against its cost, and MC-B's failure split.

signal	AUROC	extra tokens / problem
$\sigma^\star$	0.665	0
RSS	0.654	0
X1	0.666	0
B-v2 stated σ	0.655	0
MC-A raw spread, $k = 2$	0.638	496
MC-A raw spread, $k = 4$	0.710	992
MC-A raw spread, $k = 8$	0.745	1,984

A6c (development), AUROC of the signal for B-v2's run-0 answer being wrong, against the extra generated tokens per problem. MC-B's programs on the same set: 79.7 % fail at the interface, 5.1 % at run time, 14.7 % run but compute the wrong chain, 0.5 % are correct. Source: results/r5_diag/cost.json (FINDINGS #31, #33, #37).

8 Propagation of measurement noise: structure is not forecast quality

Routes that did not propagate. Three training routes failed before the recipe of Section 4.4. Supervised fine-tuning on $\sigma^\star$ itself (stage B, three seeds) taught the model to read the stated percentages, not to propagate them: its A6c slope is +0.044 [95 % CI −0.066, +0.157], and its expected score is below RSS (−0.0021 [95 % CI −0.0031, −0.0015]). σ-only RL with the predictive reward (stage C) learned to copy the smallest stated percentage, and an explicit propagation line trained on data without value-dependent derivatives (stage D) learned exponent templates rather than derivatives; both scored below RSS and X1 in their single-seed pilots. None of these training sets contained the behaviour. The text model X1 does not propagate either: its A6c slope on the frozen set is +0.002.

Implicit versus explicit. Both arms train on the same mixed-level problems; only D-pos-v2 writes its sensitivities (Section 4.4). On the frozen set, the implicit arm B-v2 learns where the noise is but not how much: its A6c slope is +0.416 [90 % CI +0.315, +0.517] (K4), short of C2's practical threshold of 0.5, and its expected score is below RSS (K8, −0.0011 [90 % CI −0.0013, −0.0009]). The explicit arm's expected score is above B-v2's (K5, +0.0013 [90 % CI +0.0008, +0.0017], Holm-rejected). Per depth, B-v2's Theil–Sen slopes (0.10–0.28) lie well below its OLS slopes (0.26–0.65): a few pairs carry the latter.

The confirmatory verdict. The primary hypothesis P is confirmed (Table 5). D-pos-v2's answer σ has an A6c slope of +0.816 [90 % CI +0.707, +0.923] (K1, Holm-rejected; per seed +0.778, +0.770, +0.900), above C2's 0.5, and an A6b slope on computation-correct pairs of +0.036 [90 % CI −0.057, +0.212], inside ±0.3 (C3). All three of its σ values propagate: the line's X (b) at +0.884 [90 % CI +0.745, +1.041] and the exact value from the line's own terms (c) at +0.843 [90 % CI +0.713, +0.987]. Problem by problem, the stated σ is close to $\sigma^\star$. Per seed, the median $|\ln(\sigma/\sigma^\star)|$ is 0.08–0.10 (RSS 0.40, X1 0.33), the slope of $\ln\sigma$ on $\ln\sigma^\star$ is 0.85–0.89 (RSS 0.48, X1 0.58), and 1σ coverage is 63 %–70 % (71 % for $\sigma^\star$ itself on the same problems).

The score secondaries, with equal prominence. Propagating noise does not buy a better expected score. Against RSS, D-pos-v2's difference is +0.0002 [90 % CI −0.0002, +0.0006] (K6, one-sided $p = 0.201$): not rejected, with per-seed estimates of both signs (−0.0005, +0.0010, +0.0001). Failure to reject is not equivalence; the interval allows a small loss or a small gain. Measured against the gap between RSS and $\sigma^\star$, D-pos-v2 closes 7 % [90 % CI −7 %, 20 %] of it. Against X1 the estimate is negative, −0.0005 [90 % CI −0.0009, −0.0001] (K7, $p = 0.991$). K7 tested whether D-pos-v2 beats X1, so we read the negative estimate descriptively only: X1 scores better. The learned σ follows $\sigma^\star$ more closely than either heuristic and yet is not shown to forecast better than either; propagating the noise correctly and forecasting well are different properties.

Depth 5, and fragile per-depth slopes. The pre-specified depth split, outside the Holm family, locates the loss. At the trained depths, D-pos-v2 is ahead of RSS (+0.0007 [90 % CI +0.0003, +0.0011]) and not distinguishable from X1 (−0.0001 [90 % CI −0.0005, +0.0003]). At the held-out depth 5 it is below both (−0.0018 [90 % CI −0.0029, −0.0008] and −0.0022 [90 % CI −0.0033, −0.0012]), while its A6c slope there is +0.820 [90 % CI +0.569, +1.054]. Depth 5 is extrapolation in chain length, one more step from the same three families, not to new physical laws. Single-depth slopes are fragile, so we give OLS and Theil–Sen side by side: from depth 1 to depth 5, the seed-mean slopes are +1.01 / +0.99, +0.84 / +1.02, +0.99 / +0.95, +0.62 / +0.89 and +0.82 / +0.93, and the number of pairs moved in the wrong direction rises from 0–3 per seed at depth 1 to 16–19 of 102 at depth 5. For one and the same adapter, the depth-5 OLS slope is +0.060 on the development set and +0.803 on the frozen set, with 20 and 19 reversed pairs; Theil–Sen gives +0.477 and +0.745. A few high-leverage reversed pairs decide the OLS slope, so we claim neither that the model propagates nor that it fails to propagate at a single depth.

Where the loss is: the tail. The regret against $\sigma^\star$ is concentrated: the worst 5 % of problems hold 69 %–83 % of each seed's expected-score regret (Table 6). Their most common primary cause is a wrong value-dependent sensitivity: a wrong formula for the derivative (43 %), or the right formula with the model's own wrong intermediates substituted (19 %). Decimal shifts in copying X into the answer (16 %) and the wrong chain (8 %) follow. The substitution errors tie the two error streams together: one wrong intermediate corrupts both μ and its sensitivity. Dropping that worst 5 % does not change the order of X1, D-pos-v2 and RSS.

Component swaps. Replacing one component of D-pos-v2's σ at a time with its exact value shows which one costs the score (Table 7). With exact sensitivities, and the model's own levels, aggregation and copy, the σ beats RSS (+0.0015 [90 % CI +0.0010, +0.0020]) and X1 (+0.0008 [90 % CI +0.0003, +0.0012]), closes 51 % of the RSS-to-$\sigma^\star$ gap, and its A6c slope rises to +0.97. Exact levels (7 % of the gap), exact aggregation (−28 %) or an exact copy (8 %) close little or none of it. This counterfactual locates the loss; it is not a method, since no model here computes

exact sensitivities. The swaps' slopes use every pair, with an unanswered member at the floor, so D-pos-v2's own row reads +0.837 rather than K1's estimate on answered pairs.

A post-hoc guardrail: the RSS fallback. A fallback that replaces D-pos-v2's σ by RSS whenever their ratio leaves [0.333, 5] (bounds chosen on the development sets and fixed before this single application) fires on 6 %–8 % of problems per seed. It lifts the expected score above RSS (+0.0016 [90 % CI +0.0015, +0.0017]) and X1 (+0.0009 [90 % CI +0.0008, +0.0010]) and closes 54 % of the gap, but it erases most of the measured propagation: the A6c slope falls to +0.136 [90 % CI −0.002, +0.267]. It is post-hoc and exploratory, it does not change K6 or K7, and a guardrail of this kind has to report both numbers.

Two black-box routes that failed: MC-A and MC-B. Executing the computation instead of predicting σ did not help either (Section 4.6). MC-A, which re-asks B-v2 on $k = 8$ perturbed copies, does not beat RSS: the robust spread's difference is −0.0062 [95 % CI −0.0088, −0.0039] (one-sided $p = 1.000$), so the pre-registered stopping rule fired and k was not increased. The model's own arithmetic errors swamp the input noise: the same perturbations pushed through the exact chain beat RSS by +0.0026 [95 % CI +0.0023, +0.0028], so the bound is attainable at this k when the computation is exact. The raw spread is a better correctness signal (Section 7), not a better σ. MC-B, in which the base model writes the problem's program, fails twice: 79.7 % of its programs fail at the interface (no program, a wrong signature, or rejected), and of the programs that run, 97 % compute the wrong chain; 0.5 % are correct. None of its pre-registered tests is rejected (0 of 5); against RSS its expected score differs by −0.0099 [95 % CI −0.0105, −0.0093].

Table 5: The R4-3 confirmatory run: D-pos-v2 and B-v2 on the frozen set.

test	statistic	estimate [90% CI]	p	Holm
K1	D-pos-v2 (a): A6c slope > 0 (primary)	$+0.816$ [$+0.707$, $+0.923$]	$<1 \times 10^{-4}$	rejected
K2	D-pos-v2 (b) line's X: A6c slope > 0	$+0.884$ [$+0.745$, $+1.041$]	$<1 \times 10^{-4}$	rejected
K3	D-pos-v2 (c) exact from terms: A6c slope > 0	$+0.843$ [$+0.713$, $+0.987$]	$<1 \times 10^{-4}$	rejected
K4	B-v2: A6c slope > 0	$+0.416$ [$+0.315$, $+0.517$]	$<1 \times 10^{-4}$	rejected
K5	D-pos-v2 − B-v2 (expected crps_log)	$+0.0013$ [$+0.0008$, $+0.0017$]	$<1 \times 10^{-4}$	rejected
K6	D-pos-v2 − RSS	$+0.0002$ [-0.0002, $+0.0006$]	0.201	–
K7	D-pos-v2 − X1	-0.0005 [-0.0009, -0.0001]	0.991	–
K8	B-v2 − RSS	-0.0011 [-0.0013, -0.0009]	1.000	–
C1–C3	D-pos-v2 (a): A6b slope $+0.036$ [-0.057, $+0.212$]	C1 yes, C2 yes, C3 yes		
C1–C3	D-pos-v2 (b): A6b slope -0.026 [-0.049, -0.007]	C1 yes, C2 yes, C3 yes		
C1–C3	D-pos-v2 (c): A6b slope -0.023 [-0.047, -0.005]	C1 yes, C2 yes, C3 yes		
C1–C3	B-v2 (a): A6b slope $+0.003$ [-0.086, $+0.129$]	C1 yes, C2 no, C3 yes		

Seed means over 3 seeds per arm, frozen R4-5 set (3,060 problems); stage B's joint pairs bootstrap, $B = 10{,}000$; 90% CIs; one-sided p; Holm over K1–K8 at 0.05. Expected-score rows: oracle-centred expected crps_log, pooled over the three sets, clustered over pairs. P = K1 ∧ C2 ∧ C3 (D125): confirmed. Source: results/stage_r4/confirmatory/results.json (D118, D125; FINDINGS #32).

Table 6: Tail autopsy: what the worst problems get wrong.

primary cause	problem-seeds	share
sensitivity: value-dependent: formula	177	39 %
sensitivity: value-dependent: substitution (own wrong intermediates)	87	19 %
copy: decimal shift	71	16 %
chain selection: wrong number of terms	35	8 %
sensitivity: power-law exponent	28	6 %
format: no answer σ	26	6 %
sensitivity: value-dependent: formula (no expression)	18	4 %
format: no parsable σ line	7	2 %
aggregation	5	1 %
sensitivity: transcription (S line right, σ-line coefficient wrong)	1	0 %
sensitivity: value-dependent: substitution (values)	1	0 %

D-pos-v2 (a), frozen R4-5 set, 3 seeds; the worst 5% of problems by regret against $\sigma^\star$ per seed (456 problem-seeds), classified by primary cause. They hold 69 %–83 % of each seed's regret. Descriptive (R5-2, D133). Source: results/r5_diag/tail.json (FINDINGS #35).

Table 7: Component swaps: which part of the explicit σ costs the score.

σ with	− RSS [90% CI]	gap closed	A6c slope
(a) as stated	+0.0002 [−0.0002, +0.0006]	7 %	+0.837
exact sensitivities (model's levels, aggregation, copy)	+0.0015 [+0.0010, +0.0020]	51 %	+0.970
exact levels (model's sensitivities, aggregation, copy)	+0.0002 [−0.0002, +0.0006]	7 %	+0.838
exact aggregation (model's terms and copy)	−0.0008 [−0.0016, −0.0001]	−28 %	+0.806
exact copy, i.e. σ = the line's X (= (b))	+0.0002 [−0.0003, +0.0007]	8 %	+0.857
model's terms, exact aggregation and copy (= (c))	−0.0018 [−0.0040, −0.0002]	−60 %	+0.826
fully exact: the builder's line, unrounded (ceiling)	+0.0030 [+0.0028, +0.0031]	100 %	+0.998

One component of D-pos-v2's σ replaced by its exact value at a time (R5-3, D133): a diagnostic of where the loss comes from, never a method. Slopes on every A6c pair, an unanswered member at the floor. RSS-to-$\sigma^\star$ gap 0.00296 (expected crps_log). Source: results/r5_diag/tail.json (FINDINGS #35).

9 Generalization: Jacobians outside the training range, and a second model

Fresh Jacobians. The J set keeps trained chains and magnitudes in range, but places one given's sensitivity outside every value that chain had in training (Section 3.5). P_J is confirmed (Table 8): D-pos-v2's J-sensitivity slope is +0.985 [90 % CI +0.868, +1.103] (J1), and its J-invariance slope is +0.026 [90 % CI −0.055, +0.119] (C3), on 72–86 computation-correct pairs per seed. The explicit arm's slope exceeds the implicit arm's by +0.530 [90 % CI +0.377, +0.676] (J2). Again there is no score gain: against RSS +0.0011 [90 % CI −0.0002, +0.0020] (J3, not rejected) and against X1 +0.0000 [90 % CI −0.0013, +0.0008] (J4, not rejected). The clamped comparator qualifies J1. A propagator that is exact but clips every sensitivity into its chain's training range already reaches +0.831 [90 % CI +0.737, +0.919], so most of the set's contrast is reachable inside the training range. D-pos-v2 exceeds it most where the dominant given was never dominant for that chain in training (+1.123 against +0.357; Table 9), but that subgroup rests on 43 pairs and is descriptive, outside the Holm family. The claim is therefore narrow: propagation through Jacobians outside the training range of trained chains. New formulas, compositions and physics were not tested.

A second model family: an exploratory pilot that returned NO-GO. We repeated both recipes on Llama-3.2-3B-Instruct with one seed, on the frozen A6c and A6b sets, with R4-3's criterion as a pre-registered go rule for running more seeds (Table 10). K1 and C2 replicate: D-pos-v2's A6c slope is +0.785 [90 % CI +0.595, +0.974] on 505 pairs. C3 cannot be measured, since only 4 A6b pairs are computation-correct, so the go rule returned NO-GO and no further seeds were run. We report this as a replication that failed at one seed, for that reason, with the same prominence a success would have had. Descriptively, outside the go rule, the A6b slope on all 510 answered pairs is −0.168 [90 % CI −0.220, −0.119]: inside ±0.3 but excluding zero, so on this model the stated relative σ shrinks slightly as the target grows, where it should stay flat (Qwen2.5-Math-1.5B-Instruct's single-seed estimate is +0.010). We claim neither scale invariance nor any other cross-model result for Llama-3.2-3B-Instruct.

Table 8: R5-6 part J: Jacobians outside the training range of trained chains.

test	statistic	estimate [90% CI]	p	Holm
J1	D-pos-v2 (a): J-sensitivity slope > 0 (primary)	+0.985 [+0.868, +1.103]	$<1 \times 10^{-4}$	rejected
J2	explicit − implicit slope	+0.530 [+0.377, +0.676]	$<1 \times 10^{-4}$	rejected
J3	D-pos-v2 − RSS (expected crps_log)	+0.0011 [−0.0002, +0.0020]	0.072	–
J4	D-pos-v2 − X1	+0.0000 [−0.0013, +0.0008]	0.442	–
J5	B-v2 − RSS	−0.0005 [−0.0008, −0.0002]	0.997	–
J6	B-v2 − X1	−0.0016 [−0.0019, −0.0013]	1.000	–
source	D-pos-v2 (a)	sens. +0.985 [+0.868, +1.103]; inv. +0.026 [−0.055, +0.119]	C1–C3: yes	
source	D-pos-v2 (b)	sens. +1.007 [+0.861, +1.172]; inv. +0.026 [−0.028, +0.078]	C1–C3: yes	
source	D-pos-v2 (c)	sens. +0.980 [+0.835, +1.143]; inv. +0.004 [−0.081, +0.082]	C1–C3: yes	
source	B-v2 (a)	sens. +0.455 [+0.348, +0.564]; inv. +0.184 [−0.032, +0.400]	C1–C3: no	
comparator	$\sigma^\star$	sens. +1.000 [+1.000, +1.000]		
comparator	RSS	sens. +0.000 [+0.000, +0.000]		
comparator	X1	sens. −0.001 [−0.003, +0.002]		
comparator	clamped to the training range	sens. +0.831 [+0.737, +0.919]		

Seed means over 3 seeds per arm on the frozen J set (1,632 problems); bootstrap as Table 5; Holm over J1–J6. C3 uses computation-correct invariance pairs. P_J = J1 ∧ C2 ∧ C3 (D147): confirmed. The clamped comparator propagates exactly but with every sensitivity clipped into its chain's training range. Source: results/r5_6/results.json (D147; FINDINGS #39).

Table 9: R5-6 part J: J-sensitivity slopes where clamping matters most.

subgroup	pairs	D-pos-v2 (a)	B-v2 (a)	clamped
exceedance below the median	408	+1.09 [+0.96, +1.22]	+0.53 [+0.37, +0.69]	+0.99 [+0.91, +1.06]
exceedance above the median	408	+0.83 [+0.63, +1.04]	+0.36 [+0.22, +0.49]	+0.62 [+0.45, +0.78]
strict	255	+0.89 [+0.65, +1.12]	+0.39 [+0.09, +0.72]	+0.71 [+0.57, +0.85]
rank flip	43	+1.12 [+0.88, +1.38]	+0.18 [−0.13, +0.41]	+0.36 [+0.00, +0.80]

Descriptive, outside the Holm family (D147 §10). Pairs: J-sensitivity and J-invariance pairs in the subgroup; slopes on the J-sensitivity pairs. Strict: also outside the range over every training chain to the same target. Rank flip: the dominant given was never dominant for that chain in training. Source: results/r5_6/results.json.

Table 10: The cross-model pilot on Llama-3.2-3B-Instruct.

item	statistic	value	met
K1, C2	D-pos-v2 (a): A6c slope	+0.785 [+0.595, +0.974] (505 pairs)	yes, yes
C3	D-pos-v2 (a): A6b slope, computation-correct pairs	4 pairs: not measurable	no
go	K1 ∧ C2 ∧ C3 (D137)		NO-GO
desc.	A6b slope, all answered pairs (Llama / Qwen s0)	−0.168 [−0.220, −0.119] / +0.010	
desc.	B-v2 (a): A6c slope	+0.396 [+0.246, +0.544]	
desc.	D-pos-v2 − B-v2 (expected crps_log)	−0.0089 [−0.0170, −0.0019]	
format	D-pos-v2: parsed / units / computation-correct	99.8 % / 99.7 % / 8.0 %	
format	B-v2: parsed / units / computation-correct	100.0 % / 100.0 % / 8.0 %	

Seed 0, frozen A6c + A6b (D137, D146). The go rule is R4-3's criterion on D-pos-v2's answer σ; C3 needs at least three computation-correct A6b pairs. Descriptive rows are D137's, outside the go rule. Sources: results/stage_xm/pilot/results.json, results/stage_xm/go_dpos2/go.json (FINDINGS #38).

10 Discussion

Structure is not forecast quality. The propagation result and the score result describe the same model. Problem by problem, D-pos-v2's σ follows $\sigma^\star$ closely (the slope of $\ln \sigma$ on $\ln \sigma^\star$ is 0.85–0.89 per seed) and moves with the source of the noise on fresh problems. Yet its expected score is not shown to beat RSS, a heuristic that ignores the values altogether, and X1 scores better descriptively. Both can hold because the score is dominated by a tail: the worst 5 % of problems carry 69 %–83 % of the regret. A σ that is right on most problems and badly wrong on a few can lose to one that is moderately wrong everywhere. The RSS fallback shows the trade directly: it buys the score, closing 54 % of the gap to $\sigma^\star$, by giving up most of the measured propagation (slope +0.14). Evaluations of stated uncertainty should therefore report a structural test (does σ move with the true uncertainty?) and a forecast score (does σ beat simple references?) side by side; either alone would have told a different story here.

Two error streams. Measurement noise is handled once the structure is right. In the component swaps, making the levels or the copy exact changes little (7 % and 8 % of the gap), and making the aggregation exact scores worse (−28 %), because on a few lines the model's own aggregate pulls an absurd root sum of squares back to a plausible size. The structure of the propagation (reading the levels, the root sum of squares, the copy into the answer) is right. What costs the forecast is reasoning error in the derivatives of value-dependent relations; exact sensitivities alone close 51 %. The two streams are not independent. When the model substitutes its own wrong intermediates, the same error corrupts both μ and its sensitivity, so on those problems the answer and its σ fail together.

Why sampling helps here and single readouts do not. On the fine-tuned model and its terse format, the tested single-completion signals (the stated σ, a linear hidden-state probe, the model's own verdict) provide little within-problem discrimination, while agreement among samples provides stronger discrimination at 8.0 times the tokens. A likely mechanism is that right answers cluster and wrong ones scatter, which only several samples can reveal. This does not show that the information is absent from the model. It may have been lost in fine-tuning, displaced to other tokens or layers, be present but not linearly readable, or be inferable only from a fuller text than the terse format provides. The exploratory comparison in Section 11 is compatible with some of these explanations and settles none of them.

Credit assignment for the tokens of σ. A proper scoring rule was not sufficient. With sequence-level credit the tokens of σ were rewarded for the answer, and the policy collapsed. Separating their credit prevented the collapse but did not teach ranking at the learning rate tested; a larger learning rate with the answer held fixed taught ranking between problems, though not within them, in a single-seed pilot. Whether that carries over to joint training of the answer and σ is open.

What would close the gap. In the counterfactual of Section 8, exact sensitivities, unlike the fallback, improve both numbers at once: the σ beats RSS and X1, and its slope rises towards one. Two routes lead there: better derivatives from the model, or an executor that computes the chain. MC-A's oracle shows that the bound is reachable when the computation is exact, and MC-B shows what stands in the way: first the interface, then the semantics of the program. An executor would

need semantic validity first, then tests of its execution, then sampling of its inputs; we leave it to future work.

11 Limitations

One model family, one scale. Every confirmatory result is on 4-bit Qwen2.5-Math-1.5B-Instruct with LoRA adapters, trained on a single laptop GPU. The replication on Llama-3.2-3B-Instruct is an exploratory single-seed pilot whose pre-registered go rule returned NO-GO: its invariance control could not be measured (4 computation-correct pairs), and its descriptive invariance slope, -0.168 [90 % CI -0.220, -0.119], excludes zero. We make no cross-model claim, and none about larger models.

A synthetic domain. The problems come from 3 formula families of mechanics; a planned held-out optics family was cut. Depth 5 is extrapolation in chain length within the same families. Part J moves the Jacobians of trained chains, not formulas, compositions or physics; the test on unseen compositions was not run before the experiment freeze. Measurement noise is relative Gaussian noise on each given, at most 5 %, and the problem text states its level; real measurements and real word problems are less tidy. The A6c construction is in-distribution for the arms trained on mixed-level data, and estimates on the development and frozen sets differ at single depths.

Single-seed and post-hoc results. Several results rest on one seed and are exploratory: E1 and its learning-rate pilots, the supervised σ student (X3), stages C and D, the first mixed-level pilot and the cross-model pilot. The comparisons with three seeds per arm (split and uniform credit, stage B, R4-3 and part J) support conclusions about these seeds, not seed-level significance. The self-consistency comparison, the tail autopsy, the component swaps and the RSS fallback were analysed post hoc, and are labelled where they appear.

Statistics and what was not run. Within-problem tests use answers sampled at $T = 1$ from the warm-up model, while the σ suffixes were trained on greedy prefixes. X1's score difference from RSS has no committed confidence interval. The originally pre-registered hypotheses on an RL-trained stated-σ model with a depth and noise curriculum (conditional calibration against post-hoc and conformal correction, decision value, stress tests) were not run, because that model was never built; the pre-registration ledger lists them as not run. Betting and other finance metrics were cut.

A post-freeze, exploratory comparison. After this paper's experiments were frozen, we read the same kind of cross-fitted hidden-state probe on the untrained Qwen2.5-Math-1.5B-Instruct, answering in its own chain-of-thought format. Within problems it discriminated more strongly there than on the fine-tuned model (0.758 against 0.539), but the same probe on a random-weights copy of the architecture, reading the same text, largely matched it (0.673). Between it and the untrained Qwen2.5-Math-7B-Instruct, the model's own verbalized verdict discriminated better at the larger size, while we observed no improvement in this probe metric between the two sizes under these conditions. The conditions are not matched (response format, training, decoding, sampler and problems all differ), so this is not evidence about what fine-tuning does to the model; it motivates a separate investigation.

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A Pre-registration ledger

[Every pre-registered hypothesis marked supported, not supported, or not run (`docs/decision_timeline.md`).]

B Reproducibility

[Reload and provenance checks, pinned bases, the bf16 merge pitfall, frozen-set checksums, compute (docs/paper_outline.md, appendix B). Every generated value, table and figure lists its committed sources in `paper/generated/SOURCES.md`.]